

POS & INVENTORY MANAGEMENT SYSTEM

Complete Enterprise Suite for Retail & Restaurant Workflows

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2.0 PROJECT OVERVIEW

1.1 Introduction

The POS & Inventory Management System is an enterprise-grade cloud business platform designed to unify retail billing and restaurant operations under a single administration panel. This platform simplifies transaction processing, regulates inventory depletion rates in real-time, coordinates supplier deliveries, and ensures strict role-based data integrity controls.

1.2 Purpose of the System

The primary purpose is to provide business owners, restaurant operators, and retail supervisors with an end-to-end management toolkit. By automating repetitive administrative tasks—such as stock deduction via recipe ingredients or billing invoicing—the platform increases overall operational efficiency and prevents manual calculation discrepancy losses.

1.3 Business Problem Statement

Modern multi-functional venues face significant friction running separate systems for retail checkouts and kitchen food orders. Data silos lead to stock counts discrepancies, raw ingredients expiring unnoticed, untracked supply chain debt, and security vulnerabilities due to cashiers holding excessive clearance privileges.

1.4 Proposed Solution

Our proposed software leverages a shared relational database to sync incoming POS checkouts and restaurant dining orders. Inventory items decrease immediately upon transaction execution. Expiries trigger automated warnings, and supply pipelines are integrated with direct PDF emailing/printing logs to prevent double-ordering or vendor misunderstandings.

3.0 SYSTEM OVERVIEW

2.1 Core Concepts

The platform bridges POS billing, restaurant operations, supplier coordination, and user roles. Staff log into dedicated viewports (Cashier POS register, Kitchen KOT grid, Admin settings). Transactions process instantly, synchronizing changes across the system.

2.2 Business Benefits

- Real-time stock visibility across multiple branches and kitchen shelves.
- Leakage prevention via receipt audit logs and transaction validation.
- Immediate customer engagement through WhatsApp-delivered invoices.
- Improved vendor relationships with structured purchase order templates and print audits.

2.3 Multi-Business Support

The system is architected to support standard retail shops, hybrid cafes, and full-service dining tables. Super Admins can configure custom parameters (GSTIN, Service Charge rates, SMTP credentials, digital signatures, and official seals) to conform with local tax laws and branding specifications.

4.0 USER ROLES & ACCESS MANAGEMENT

The system implements strict Role-Based Access Control (RBAC) to ensure that staff can only execute commands matching their operational scope.

Role: Super Admin

Full system control, business onboarding, database configuration, settings overrides.

Role: Admin

Configures inventory, registers suppliers, approves stock requests, views full analytics reports.

Role: Manager

Manages daily cash deposits, resolves expiring stock notices, edits products catalog, adjusts table maps.

Role: Cashier

Operates POS Billing screen, handles returns/exchanges, opens/closes cash drawers.

Role: Kitchen Staff

Views live KOT tickets, updates prep status, submits stock requests for raw ingredients.

Role: Waiter

Takes table orders, assigns tables, logs guest reservations, manages waiter dashboard tasks.

Role: Inventory Staff

Receives goods (GRN entry), updates inventory stock counts, registers damages/wastes.

5.0 LOGIN & AUTHENTICATION MODULE

Authentication represents the security perimeter of the application. Every user interaction is validated against a cryptographically secure session context.

4.1 Token Security & Encryption

All user passwords undergo high-cost hashing using bcrypt before storage. Upon successful credentials match, the server generates a JSON Web Token (JWT) signed with a secret key. This token is transmitted via HttpOnly cookies to block Cross-Site Scripting (XSS) extraction.

4.2 Session Control

Sessions remain active for 8 hours. The system checks token expiry on every request. If a session is idle, or if the token has expired, the user is redirected to the login screen. Session logs store IP metadata and browser user-agents to audit logins.

4.3 Forgot Password Flow

If a user loses credentials, they can request a password reset. The system generates a single-use cryptographically random token with a 15-minute expiration, sent directly to their email. Reset tokens are instantly flagged as invalid after a single use.

6.0 DASHBOARD MODULE

5.1 Purpose & Layout

The dashboard displays business performance metrics. It provides real-time statistics to assist operators in making informed decisions.

5.2 KPI Metrics & Real-time Charts

- Sales Today: Total completed checkout revenue.
- Active Orders: Real-time KOT tickets and held bill counts.
- Low Stock warning count: Instant alert on product shortages.
- Expiry notices count: Alerts for batches expiring within 30 days.

5.3 System Alerts & Notifications

The notification sidebar broadcasts system events immediately—such as a kitchen stock request pending approval, a waiter calling for bill settlement, or stock levels dropping below threshold limits.

7.0 RETAIL MANAGEMENT MODULE

6.1 Product & Category Control

Products are indexed with parameters (SKU, Category, Brand, Purchase Price, Selling Price, and current stock counts). Categories support parent-child nesting structures (e.g., Electronics > Mobile Phones) to catalog stock logically.

6.2 Stock Tracking & Barcode Generation

Cashiers scan barcodes at checkout, which automatically fetches the product from the database. The system includes built-in PDF barcode sheet generators to tag unlabeled inventory.

6.3 Customer Relationship Management

Store operators can log customer demographics and loyalty points. Every purchase increments points, which customers can redeem as discounts during future checkouts.

6.4 Returns, Exchanges & Refunds

When products are returned, cashiers log the reason (e.g., damaged, incorrect size). The system registers the items back into inventory (or marks them as waste), adjusts sales ledgers, and generates a store credit invoice.

8.0 RESTAURANT MANAGEMENT MODULE

7.1 Interactive Table Management

The digital table layout shows live occupancy statuses (Available, Occupied, Cooking, Ready, Served, Cleaning). Waiters can click tables to place orders, split checks, or request bill settlements.

7.2 QR Menu Ordering System

Each table features a unique QR code. Guests scan the QR code to load a digital menu builder page. They can place orders and complete checkout payments via simulated UPI/Razorpay gateways.

7.3 Kitchen Display System (KDS)

Kitchen screens display active tickets. Color-coded cards show preparation timers to flag delayed tickets. Updates on KDS instantly notify waitstaff.

7.4 Waitstaff Dashboard & Tasks

Waiters access a tablet view showing assigned tables, ready KOT notifications, and service requests. This reduces wait times and improves table turnover rates.

9.0 INVENTORY MANAGEMENT MODULE

8.1 Product Catalog

Maintains centralized control over all inventory. Stock modifications from POS sales, KOT recipe deductions, or supplier check-ins update this catalog instantly.

8.2 Low Stock & Out of Stock Alerts

Each item includes a minimum stock warning threshold. When stock falls below this level, the system triggers alerts and auto-generates purchase request drafts.

8.3 Smart Expiry Alerts & Resolution Actions

Batches log expiry dates. The expiry dashboard filters stock into risk buckets (30 days, 15 days, 7 days, Expired). Managers can resolve items via markdown promotions or stock transfers.

8.4 Inventory Activity Logs

Every inventory adjustment (sales, purchase check-in, waste log, recipe deduction) writes a record containing timestamp, change amount, operator, and reason, ensuring auditability.

10.0 SUPPLIER MANAGEMENT MODULE

9.1 Supplier Directory & Mapping

Tracks contact records, tax numbers, and pricing agreements. Mappings tie specific inventory items to suppliers, simplifying reordering.

9.2 Purchase Order lifecycle

Admins draft purchase orders (PO), submit them for approval, and track delivery. Upon receipt, a Goods Received Note (GRN) verifies quantities, registers damages, and adds items to inventory.

9.3 Printing Audit Logs & Custom Settings

POs log printing events (timestamp, username, print count) to prevent duplicate printout exploits. Custom SMTP configurations allow emailing PO PDFs directly to suppliers.

11.0 KITCHEN REQUEST & PROCUREMENT MODULE

This module manages stock replenishment by connecting kitchen staff directly with suppliers and administrators.

10.1 Procurement Workflow

1. Kitchen Staff: Creates a stock request for low ingredients.
2. Admin Approval: Admin reviews requests in the Supplier dashboard.
3. Purchase Order: Approved requests compile into a PO.
4. Vendor Contact: System generates PO PDF and emails/whatsapps the vendor.
5. Delivery: Warehouse audits delivery and updates inventory counts.

10.2 Workflow Advantages

This workflow prevents verbal-order errors, traces procurement delays to specific parties, and ensures all updates match authorized purchase requisitions.

12.0 POS BILLING MODULE

11.1 Real-time Cashier Interface

Designed for fast checkouts, the interface features barcode scanner listeners, category filters, and quick-add grids to build carts quickly.

11.2 Calculations, Taxes & Coupon Discounts

The checkout engine calculates subtotal, discounts, and tax rates. Supported payment methods include Cash, Card, UPI, and split distributions.

11.3 Invoice Generation

On checkout completion, the system saves the sale, generates a transaction record, creates a PDF invoice, and prints a copy.

13.0 INVOICE MANAGEMENT MODULE

12.1 Purpose & Compliance

Centralizes invoice records to ensure auditability and compliance with local tax guidelines.

12.2 Invoice Lifecycle

- Draft: Held bills awaiting checkout confirmation.
- Completed: Settled invoices with PDF receipts.
- Cancelled: Voided checkouts requiring administrative overrides.

12.3 Digital Sharing Integrations

Includes direct API endpoints to email PDF receipts or send them to customer mobile numbers via WhatsApp notifications.

14.0 CUSTOMER MANAGEMENT MODULE

13.1 Database Records

Tracks customer profiles, including emails, mobile numbers, DOB, locations, spend histories, and active loyalty balances.

13.2 Loyalty Program Logic

The system converts transaction values into loyalty points. Rules define point conversion and validation limits, protecting store profit margins.

15.0 REPORTS & ANALYTICS MODULE

14.1 Business Intelligence Sales Reports

Consolidates sales data into hourly, daily, and weekly charts. Filters group revenue by cashiers, payment methods, and categories.

14.2 Supplier Performance Reports

Tracks supplier delivery speeds, damaged item rates, and ledger payments to evaluate vendor reliability.

14.3 Profitability Audits

Subtracts purchase costs from transaction values to calculate net margins. Expiry losses and damaged items are factored in to show true profitability.

16.0 WHATSAPP INTEGRATION MODULE

15.1 Direct Customer Outreach

Sends transactional notifications (e.g., invoices, loyalty updates) directly to customer mobile numbers.

15.2 Supply Chain Notifications

Integrates with the supplier module to dispatch PO PDF files to vendor mobile numbers, cutting down on email processing delays.

17.0 EMAIL INTEGRATION MODULE

16.1 System Notifications

Sends system notifications, reset password tokens, and critical system updates to staff and administrators.

16.2 Vendor PO PDF Emailing

Generates purchase order PDFs and emails them directly to suppliers. Configurable SMTP settings allow businesses to send emails from their own domains.

18.0 DATA FLOW DIAGRAMS

17.1 User Login DFD

User -> Submit Credentials -> Auth Middleware -> Validate Hash -> Generate JWT -> Access Dashboard

17.2 Retail Checkout DFD

Cashier -> Scan Barcode -> Fetch Product -> Settle Bill -> Print Invoice -> Decrement Stock Count

17.3 Kitchen Order DFD

Guest QR -> Submit Order -> Kitchen Display -> Prepare Food -> Waiter Serve -> Settle Bill

17.4 Supplier Procurement DFD

Low Stock Warning -> Generate PO -> Admin Approval -> Email Vendor -> Receive Delivery -> Update Inventory

19.0 SYSTEM ARCHITECTURE

18.1 Tech Stack Components

- Frontend: Single Page Application built on React, TypeScript, and Tailwind CSS.
- Backend: REST API Server running Node.js, Express, and TypeScript.
- Database ORM: Prisma Client interacting with PostgreSQL.
- Communications: WebSockets / Server-Sent Events (SSE) for live KOT sync.

18.2 API Gateway Layer

Acts as the entry point, resolving routing, JWT validations, role authorization checks, and rate-limiting limits.

20.0 DATABASE OVERVIEW

The database uses a relational schema designed to ensure data consistency and referential integrity.

Table: User

Stores authentication credentials, contact info, status flags, and role assignments.

Table: Product

Tracks pricing, SKU indexes, expiration dates, statuses, and supplier mappings.

Table: Order / OrderItem

Stores completed checkouts, payments, discounts, cashier details, and purchase list histories.

Table: RestaurantTable

Logs seating capacities, table numbers, and active order IDs.

Table: KitchenOrder

Maintains KOT ticket items, waiters, preparation timestamps, and table numbers.

Table: Supplier

Tracks vendor contacts, tax numbers, purchase history logs, and rating points.

21.0 SECURITY FEATURES

20.1 Password Cryptography

Passwords undergo high-cost hashing using bcrypt. The system never stores plain-text credentials, protecting accounts against database leaks.

20.2 Route Protections

Route middleware blocks requests that lack signed authorization tokens. APIs restrict access based on user roles (e.g., only Admin/Super-Admin can access SMTP configurations).

20.3 Database SQL Injection Protection

Prisma ORM parameterizes database queries automatically. This protects the backend against malicious SQL injections.

22.0 PROJECT BENEFITS

21.1 For Business Owners

Consolidates retail sales, menu mapping, supplier orders, and expiries under one dashboard. This reduces manual workload, lowers staff costs, and prevents inventory leakage.

21.2 For Kitchen & Waitstaff

Automates order handovers, alerts waiters when food is ready, and updates stock levels dynamically. This minimizes preparation delays and table check errors.

21.3 For Customers

Guests enjoy QR-based table ordering, digital payments, and digital receipts sent to their phones via WhatsApp/Email.

23.0 CURRENT PROJECT STATUS

The platform is actively running in staging and development environments. Core checkout flows, inventory tracking, KOT preparation logs, and supplier emailing are certified.

[SCREENSHOT PLACEHOLDER]

Screen 1: Dashboard Analytics Viewport

Summary cards displaying daily profits, active dining orders, and low-stock alerts.

[SCREENSHOT PLACEHOLDER]

Screen 2: Inventory Catalog & Expiry Center

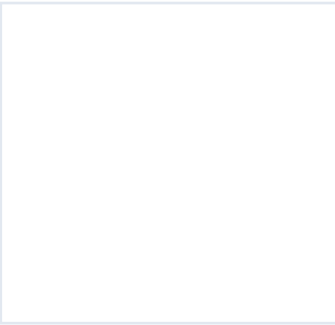
List view of all stock items, color-coded expiration categories, and quick resolution controls.

[SCREENSHOT PLACEHOLDER]

Screen 3: Cashier POS billing checkout

Real-time billing viewport with barcode scanner listener and split-

payment sliders.



24.0 FUTURE ENHANCEMENTS

23.1 Mobile App Releases

We plan to release native iOS and Android applications for restaurant waiters, cashiers, and managers to simplify remote administration.

23.2 Multi-branch Franchise sync

We are developing multi-tenant franchise synchronization. This will enable managers to sync stock levels and menus across regional outlets from one central dashboard.

23.3 AI-Powered Predictive Sourcing

We will integrate machine learning algorithms to analyze historical sales data. The system will predict ingredient demands and automatically schedule supplier orders to minimize waste.

25.0 CONCLUSION

The POS & Inventory Management System is a comprehensive, production-ready enterprise business management platform. It streamlines retail transactions, recipe deductions, expiries, and supplier orders.

By unifying dining workflows, retail checkouts, and supply chains into a shared relational database, the platform eliminates administrative silos and protects margins.

With its robust security controls, real-time alerts, and built-in SMTP integrations, this platform is ready for deployment across retail and hospitality venues.
